

USER GUIDE & TROUBLESHOOTING

HPC Whisper

Type on a Windows CE handheld.
Read it on your phone.

VERSION 1.2



18 – 21.5 kHz
near-ultrasound

No cable
no network

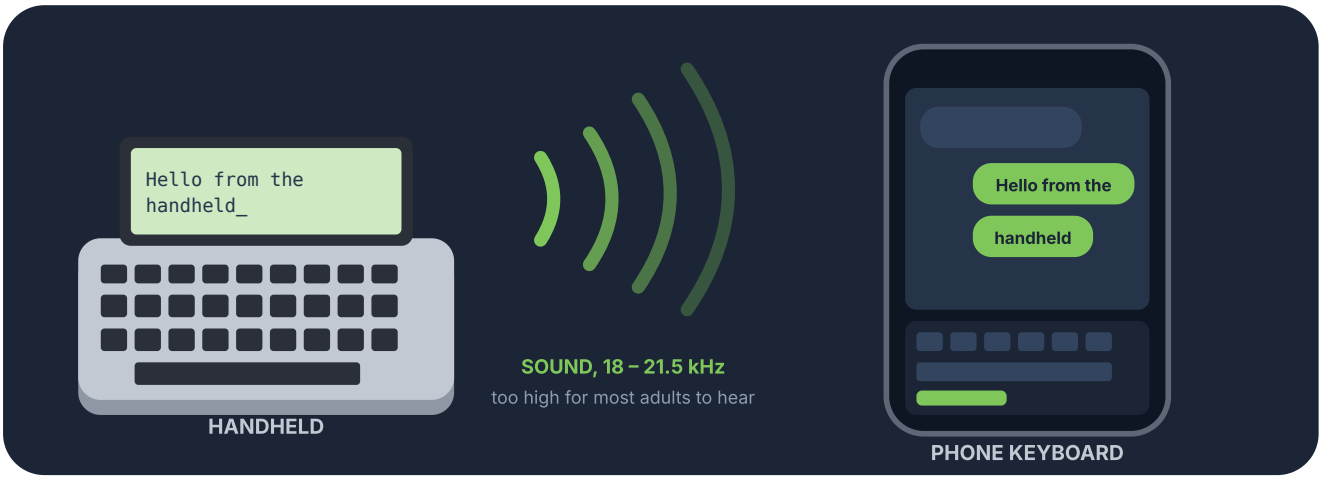
Freeware
no ads

Made on and for the HP Jornada 720 · github.com/kiriakoschristodoulou-netizen/hpc-whisper

AT A GLANCE

Your handheld gets a voice. Your phone gets the words.

Type on the handheld and the text appears in any app on your Android phone, carried by sound too high for most people to hear. No cable, no Bluetooth, no network, no extra hardware.



What you need

A handheld

ARM-based Windows CE 3.0 or later. Tested on the **HP Jornada 720**. Not for MIPS or SH3 machines (Jornada 680, 690).

An Android phone

Android 7.0 or later, with a microphone. On an iPhone use the receiver web page (Copy and Share); Apple blocks microphone keyboards.

A calm room

Phone and handheld within about **one metre**. Some chargers and ultrasonic devices make noise at 18–21 kHz and can disturb it.

Set up in five minutes

1. **Install the phone app**
   Open `HPC-Whisper.apk` from the latest release.
2. **Copy the handheld folder**
   Unzip `HPC-Whisper-12.zip` and copy the folder to the handheld.
3. **Switch the phone keyboard**
   In any text field, choose the HPC Whisper keyboard.
4. **Run and type**
   Start `HPCWhisper.exe`, hold the phone near the speaker.

Get it from one place. Download the apps only from the official release page of the project. The phone app is signed with a key that is public on purpose, so look-alikes are possible elsewhere.

PART 1

Set up the phone

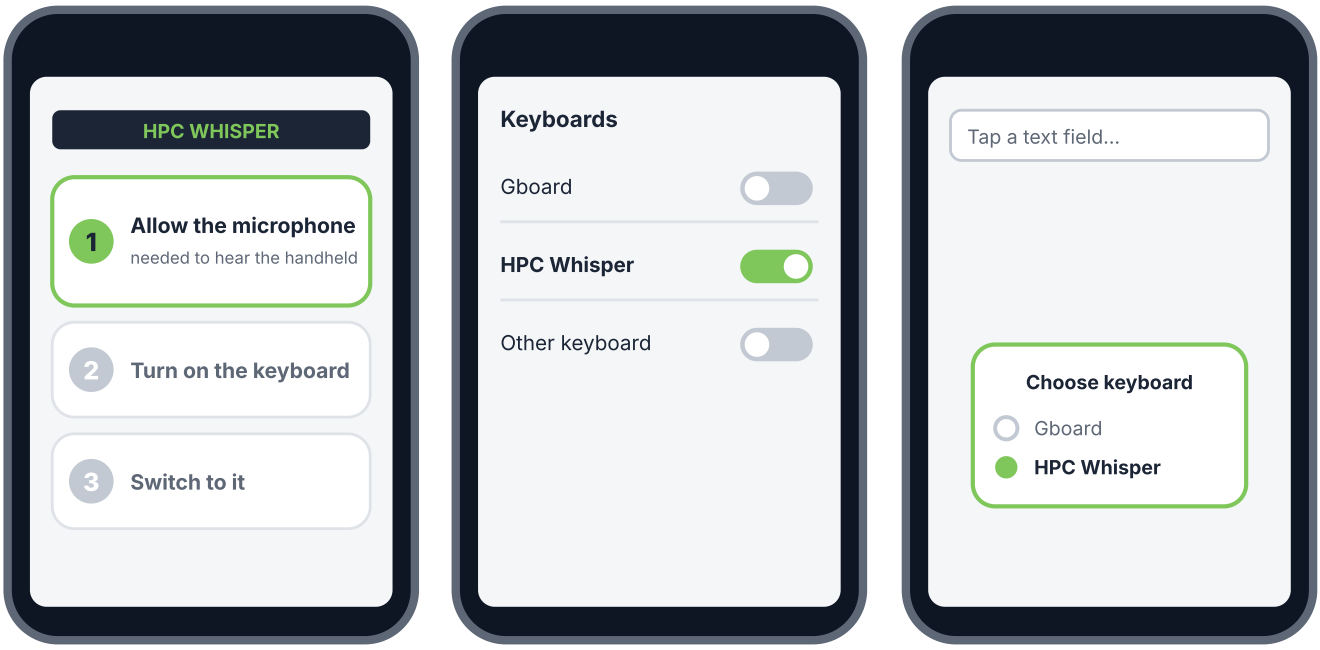
Install the app, then tap each of the three steps it shows. It takes about a minute.

1 Install the app

Download `HPC-Whisper.apk` from the latest release and open it. Android asks once to allow installing from this source.

2 Open HPC Whisper and tap the three steps

The app lists them in order. Tap each one.



Allow the microphone. It listens only while this keyboard is open.

Turn on the HPC Whisper keyboard in the list that opens.

Switch to it with the keyboard picker, in any app.

Schematic drawings; your phone's screens look a little different.

Optional: Send, Back, Home

Turn on the **HPC Whisper helper** in the phone's accessibility settings. It lets the handheld press **Send**, **Back**, **Home** and **Recent apps** on the phone. It reads nothing and stores nothing, and it is off until you turn it on.

The keyboard on the phone

The keyboard has touch edit keys: arrows, select, copy, paste and undo, so you can correct text without touching the handheld.

Everything that arrives is kept on the phone as **dated sessions** (the newest 25 by default, never fewer than 10), hidden until you tap them. You can turn this off in **Settings, Saved sessions**.

Status words. **"Waiting for a missing piece"** means a piece is on its way: wait a few seconds. **"Weak signal"** means move the phone closer to the handheld's speaker.

PART 2

Set up the handheld

Three moves, and you are typing.

1. **Download and unzip**
   Take `HPC-Whisper-12.zip` from the same release and unzip it on a computer.

2. **Copy the folder**
   Copy the folder `HPC-Whisper` to the handheld. A storage card or the main memory both work.

3. **Run it**
   Start `HPCWhisper.exe`. Hold the phone within a metre of the handheld's speaker, microphone side toward the handheld, not in a pocket or face-down on a soft surface.

Keys

Type as usual: everything goes to the phone as you type. The program's own shortcuts use `Alt`, so every `Ctrl` key reaches the phone.

TYPING AND MOVING	
Letters, Enter	Typed into the field
Arrows, Home, End, PgUp, PgDn	Move. Shift selects; Ctrl jumps
Backspace, Del	Delete. Ctrl deletes a word
Tab, Shift+Tab, Esc	As on a computer
Ctrl+A, C, X, V	Select all, copy, cut, paste
Ctrl+Z, Ctrl+Y	Undo, redo

PROGRAM SHORTCUTS	
`Alt` + `S`	Press **Send** in the app
`Alt` + `B`, `H`, `W`	Back, Home, Recent apps (helper)
`Alt` + `R`	Send the last text again
`Alt` + `N`	New page: the phone starts fresh
`Alt` + `K`	Keys list
`Alt` + `O`	Settings
`Alt` + `Q`	Quit

Sending a message in a chat app

Enter normally gives a new line. To **send** instead:

OPTION A **Press Alt+S**
Or tap Send on the handheld's screen. Works any time.

OPTION B **Make Enter send**
In **Settings, Enter key** choose "Sends the message". Enter then sends, and `Shift` + `Enter` is a new line.

Needs one of: the helper turned on in the phone's accessibility settings, **or** a text field whose own key says Send, Go or Search (the keyboard then uses that key).

PART 3

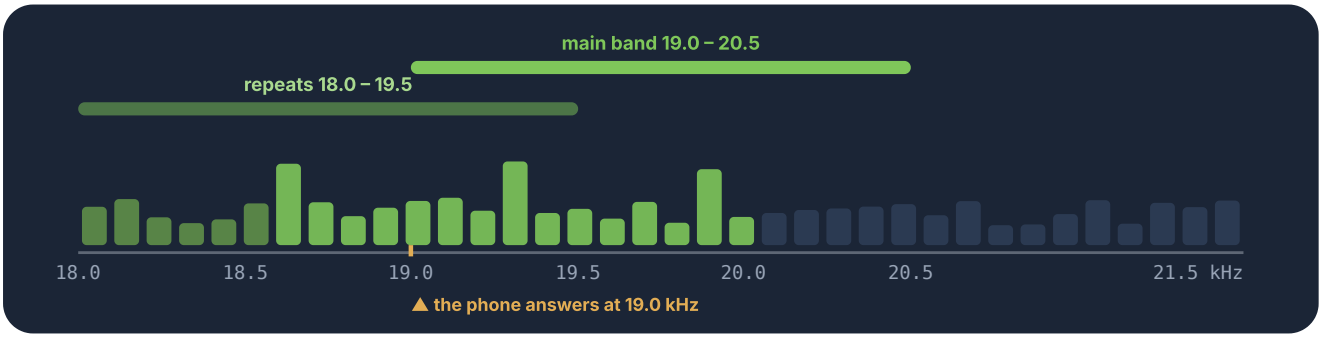
Handheld settings

Press `Alt` + `0`. The defaults suit most handhelds; change one thing at a time.

SETTING	WHAT IT DOES	DEFAULT
Repeats	How many times each piece of text is sent (2 to 8). More is surer.	**6**
Second band for repeats	Every second send goes one kHz lower, where a weak spot at the first band does not reach it.	**On**
Volume	Loudness of the tones.	**50 %**
Speed	Normal is the surest; Fast shortens the wait for the text to appear.	**Fast**
Band	The phone finds the band by itself; change this only if something else makes sound up there.	**19.0 – 20.5 kHz**
Sound rate	The Jornada's own 43,200 or Windows' 44,100. Leave it at 43,200; try 44,100 only if the sound or the microphone will not open.	**43,200**
Return channel	The phone confirms what it received, silently, so nothing is repeated needlessly. Turn on *Confirm to the handheld* on the phone too.	**Off**
Enter key	New line, or "Sends the message" (Shift+Enter is always a new line).	**New line**
Test	Sends a sentence to the phone, to try it.	
About	Version and credits.	

Where the sound lives

The text is sent as a comb of quiet tones between 18 and 21.5 kHz. Each piece is repeated, alternately on the main band and a second band a kilohertz lower, so a weak spot at one frequency does not lose it.

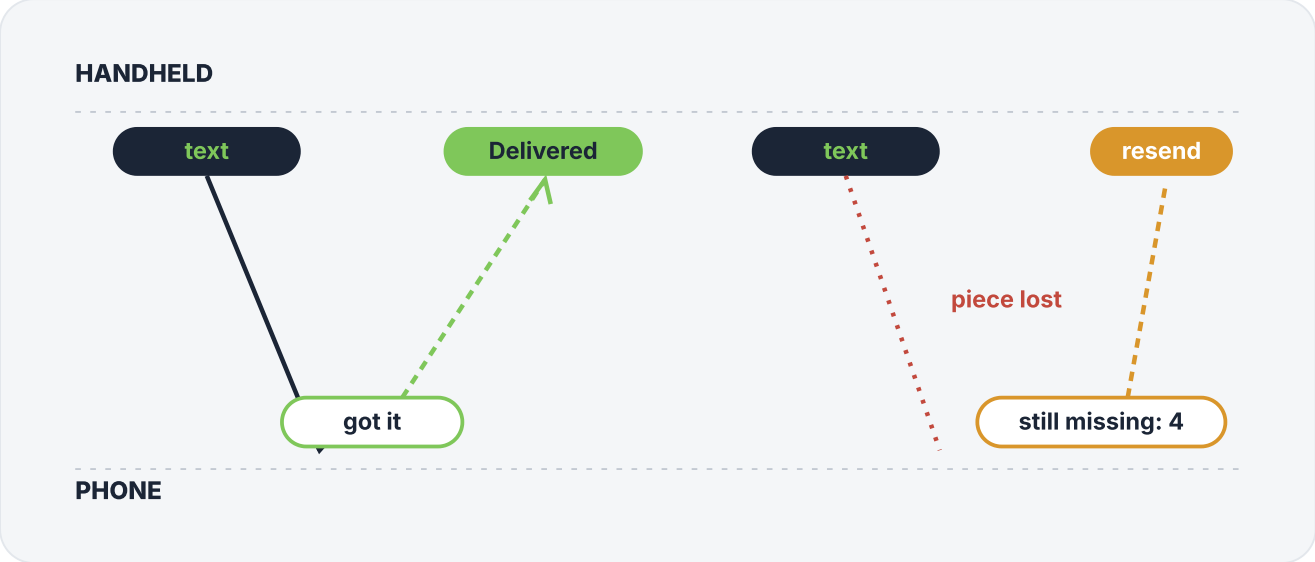


Why near-ultrasound? It is quiet to most people, it travels well through air over a metre, and a phone's microphone and a handheld's speaker both reach it. The text itself is protected by error correction and a checksum, so wrong letters are rare.

PART 4 · OPTIONAL

The return channel

Normally the handheld just repeats everything. With the return channel on, the phone quietly answers, so the handheld knows what arrived.



Arrived? Stop.

Text that has arrived is not sent again, and the title bar shows **Delivered**.

Missing? Resend.

A missing piece is sent again at once instead of waiting for the next repeat.

Check every 30 s

When you stop typing, the handheld asks what the phone holds. If it differs anywhere, the whole page is sent again.

Turn it on (both sides)

H **Handheld**

`Alt` + `O` → **Return channel** → On.

P **Phone**

Settings → **Confirm to the handheld** → On. Keep the keyboard open and the media volume up.

The keyboard also fixes text it had already written into the field if the page changed after that.

What the handheld's Return channel row tells you

On, listening for the phone	Switched on; no answer from the phone yet.
On, the phone confirms	Answers are arriving. All is well.
On, but …	The handheld's microphone could not be used; the reason follows. Everything still works as with the channel off.

Only one side on? Nothing breaks: the handheld simply repeats as set under Repeats. The receiver web page (iPhone and others) cannot answer, so it always works that way.

PART 5

When something isn't right

Find what you see on the left; try the fix on the right. The first three solve most problems: the phone closer, the right keyboard, the volume up.

On the phone

Nothing appears on the phone	The HPC Whisper keyboard must be the one in use (switch with the keyboard picker) and the microphone allowed. Hold the phone within a metre of the handheld's speaker. Raise the handheld's **Volume** in Settings.
"Weak signal"	Move the phone closer, microphone toward the speaker. Remove a case that covers the microphone. Try **Settings, Speed: Normal**.
"Waiting for a missing piece"	Wait a few seconds: it fills in by itself. If not, press `Alt`+`R` on the handheld. A piece still missing after half a minute is given up, so it cannot hold up the text after it.
Some text is missing or left over	Press `Alt`+`R`. Turn on the return channel on **both** sides: it checks the whole page every 30 s. If it keeps happening, report it with the phone's log (see "Sending the log").
A wrong letter appears now and then	Press `Alt`+`R`. If it happens often, move closer, reduce background noise and try Speed: Normal.

Sending and phone keys

Enter does not send a message	Set **Settings, Enter key** to "Sends the message" and turn on the helper in the phone's accessibility settings, or use `Alt`+`S`. The app's field may have no Send key.
Send / Back / Home do nothing	They need the **helper**, which is turned on in the phone's accessibility settings.

On the handheld

"No sound device found"	The sound device could not be opened. Close other programs that play or record sound and start again.
"The sound device would not open at … samples/s"	**Settings, Sound rate:** try 44,100.
It does not run on my handheld	It needs an **ARM** processor and Windows CE 3.0 or later. MIPS and SH3 machines (Jornada 680 and 690 and others) are not supported.

PART 5, CONTINUED

Return channel, sound, reports

Return channel

"On, but No microphone found"	This handheld has no microphone it can use. Everything still works as with the return channel off.
"On, but The microphone would not open"	Another program may have the microphone: close it and restart HPC Whisper. If it persists, try **Settings, Sound rate: 44,100**.
"On, listening for the phone" for a long time	The phone is not answering. Turn on **Settings, Confirm to the handheld** in the phone app, keep the keyboard open, the phone close and its **media volume** up.
The handheld never shows "Delivered"	Delivered appears only with the return channel on, **on both sides**. Check that Confirm to the handheld is on in the phone app, the keyboard is open, the phone is close and its media volume is up. Without answers the handheld simply repeats as set.

Sound

The sound is audible or annoying	Lower **Volume** in Settings. Some people, children and pets hear the top of the range.

Sending the phone's log

1. In the phone app open **Settings → Troubleshooting**.
2. Tap **Share the log** right after the problem: that is the best time.
3. Send the zip together with the bug report below.

Please note
The log holds a text record of what the phone heard and the last 30 seconds of sound. That may include conversation in the room, so share it only if you are comfortable with that.

A good bug report

TELL US	WHY IT HELPS
Handheld model and Windows CE version	Sound chips differ between machines.
Phone make, model and Android version	Microphones and filters differ too.
Return channel on or off, both sides	The two modes behave differently.
What you typed, what appeared, what the status said	Shows where text was lost.
The phone's log, if you are comfortable sharing it	Lets us hear what the phone heard.

PART 6

Privacy & safety

What the apps do

No internet

The Android app declares **no network permission**. Nothing you type is sent to any server.

Microphone only when needed

The phone listens only while the HPC Whisper keyboard is open, and only to decode the handheld's sound. The log's 30 seconds of sound stay on the phone until you clear or share them.

Saved sessions

What arrives is kept as dated sessions. Switch it off in **Settings, Saved sessions**, or delete them.

The helper

An accessibility service, off until you turn it on. It only does what the handheld asks (Send, Back, Home, Recent apps), reads nothing and keeps nothing.

Keep in mind

Sound is not private

The text travels through the air. Anyone close enough with a microphone and the published format could decode it. **Do not type passwords** or anything secret.

Who hears it?

The tones are about 18–21.5 kHz. Most adults cannot hear them; some young people, dogs and cats can. Lower **Volume** if it bothers anyone.

Volume

The sound is not harmful at the volume of a handheld's small speaker. Keep it modest anyway. For the return channel, keep the phone's **media volume** up: the phone plays a short confirmation sound that most people won't notice.

Permissions

The microphone, and vibration only for the optional "Buzz when text arrives" (off by default). The handheld stores only its settings, in a small file next to the program.

Official downloads only. The Android app is signed with a key that is in the repository on purpose, so new builds install over old ones. That means anyone can build an app that looks the same: **get it only from the official release page**.

Freeware

HPC Whisper is freeware: free to use, free to pass on complete and unchanged, not to be sold, no warranty. The full text is in `LICENSE.txt`. If you enjoy it, there is an optional tip link in the app; it is entirely up to you.



Thanks for trying HPC Whisper.

Tell us which handhelds and phones you tried it on and what happened. Every report, good or bad, helps.

`github.com/kiriakoschristodoulou-netizen/hpc-whisper`